

DSP Midterm, Group 03

Speech and silence segmentation

Võ Thanh Quân: Binary Search

Vương Quốc Trung: Histogram

Đinh Huỳnh Nguyên Khang: Statistics

Shared experiment settings

4 TRAIN WAVs. 4 separate TEST WAVs.

25 ms windows centered on 10 ms decisions.

Normalize STE separately for each WAV.

Merge all silence shorter than 200 ms.

Red: LAB. Blue: predicted boundaries.

F0 is shown as an independent estimate.

Binary Search

Võ Thanh Quân

TRAIN

Normalize STE
Apply median

LAB frame labels
Keep overlap

Bisect threshold
Score TRAIN

Median = 15. Shared T = 0.0016054242.

TEST

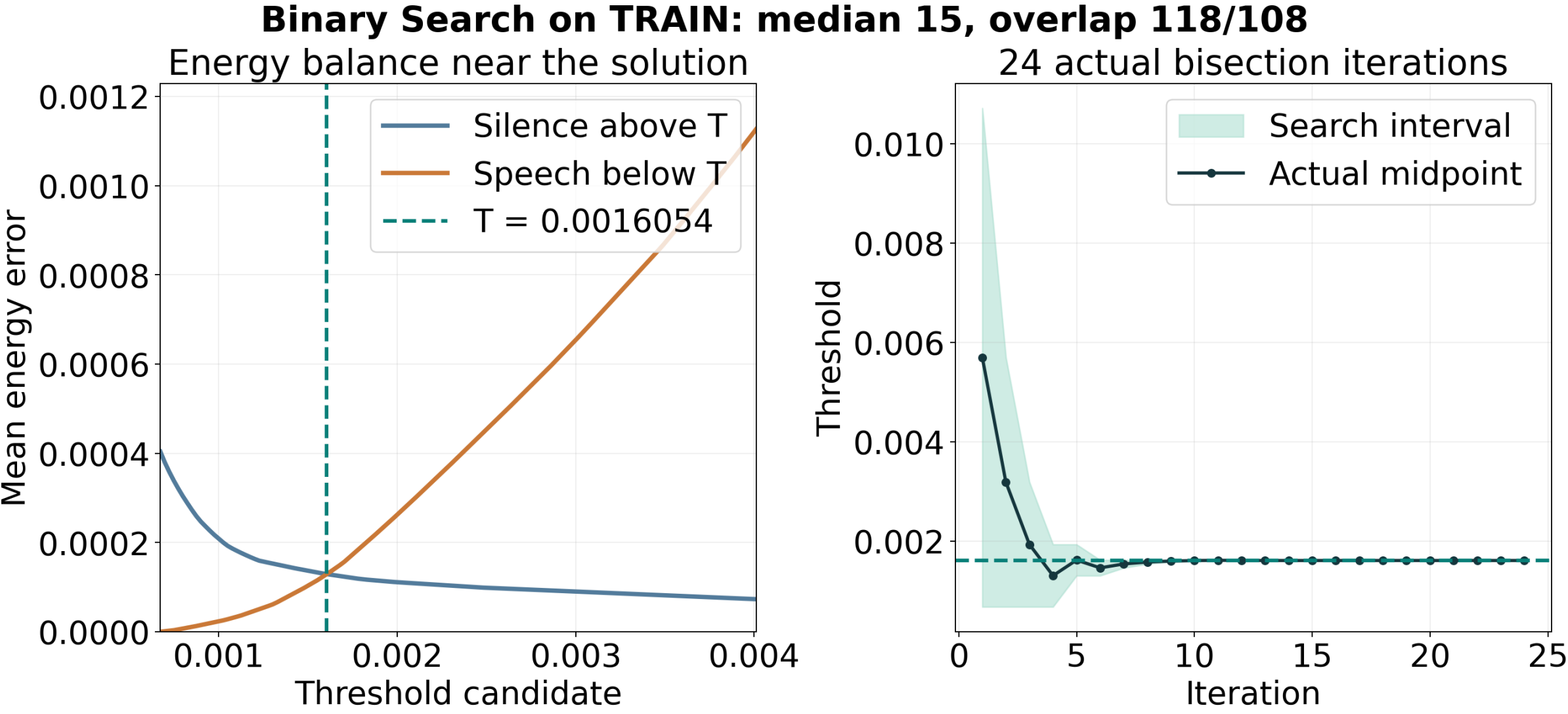
Same TEST features
Median = 15

Fixed threshold
Speech or silence

Fill short silence
Under 200 ms

Binary Search: finding the balance

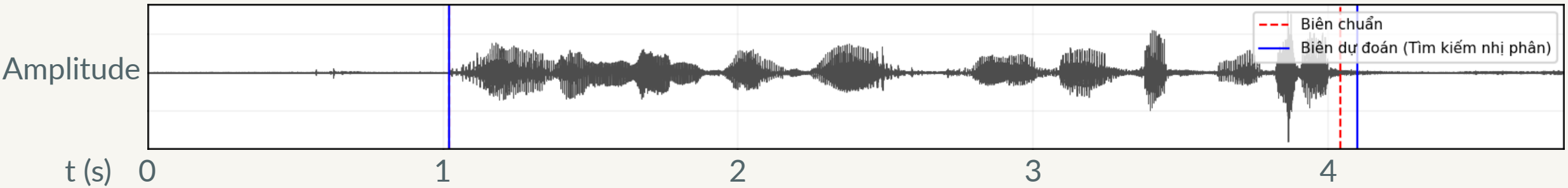
TRAIN overlap. 24 actual bisection iterations.



Binary Search: phone_F2

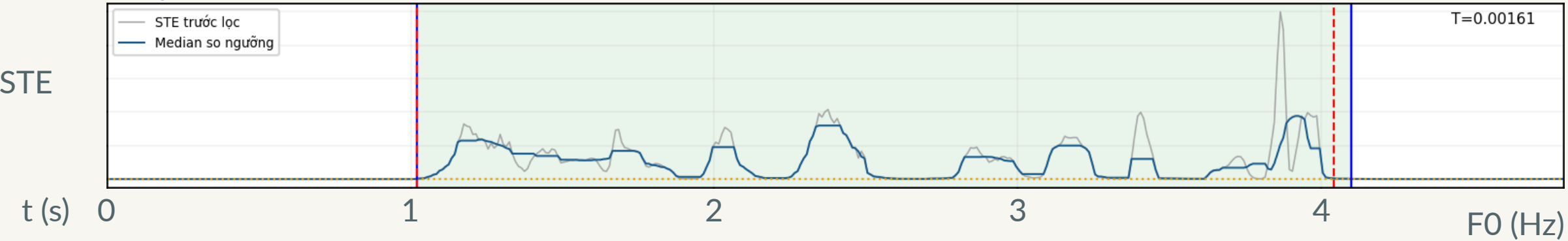
MAE 30 ms

Start matches. End 60 ms late.



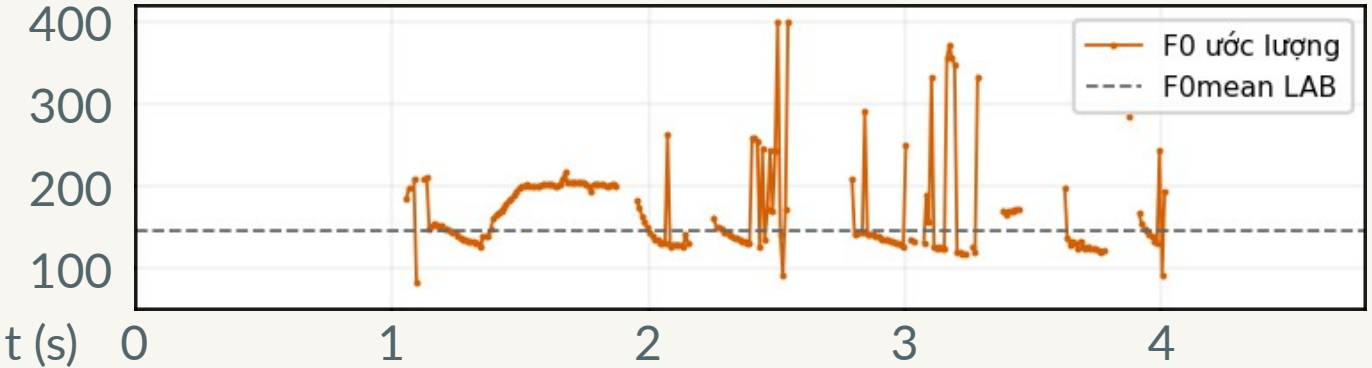
Grey: raw STE. Blue: median STE.

T = 0.0016054242



Tail median STE remains above the fixed threshold

Extra: 0. Missed: 0.



Binary Search: phone_M2

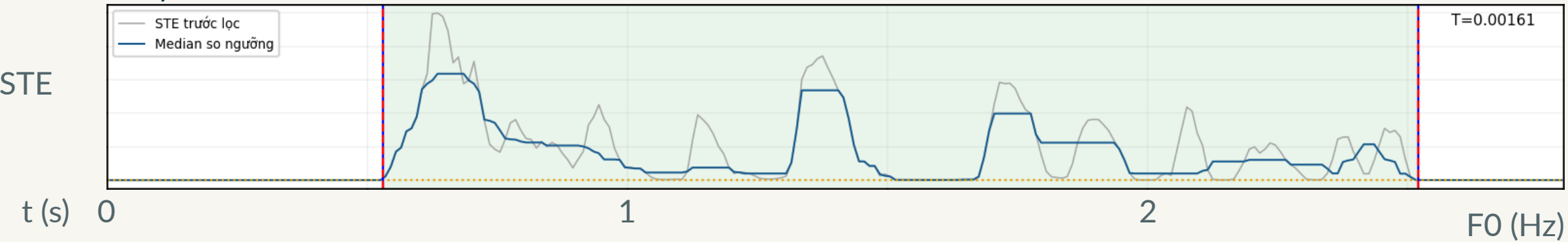
MAE 0 ms

Start matches. End matches.



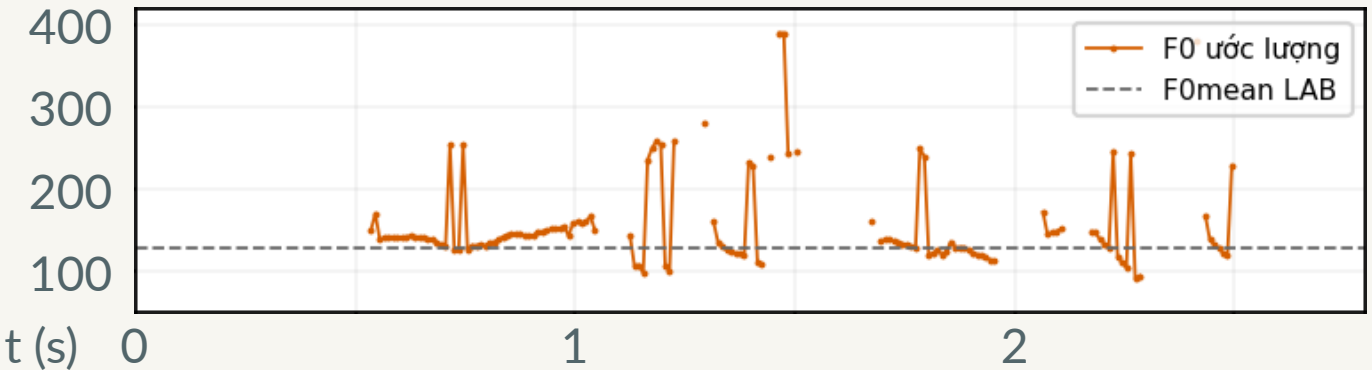
Grey: raw STE. Blue: median STE.

T = 0.0016054242



Both LAB timestamps match
the detected transitions

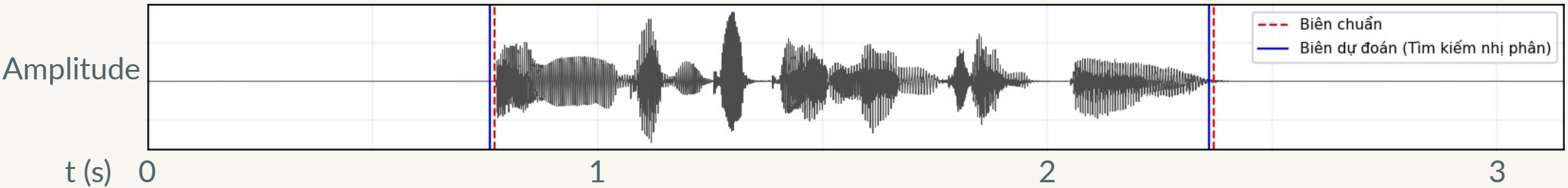
Extra: 0. Missed: 0.



Binary Search: studio_F2

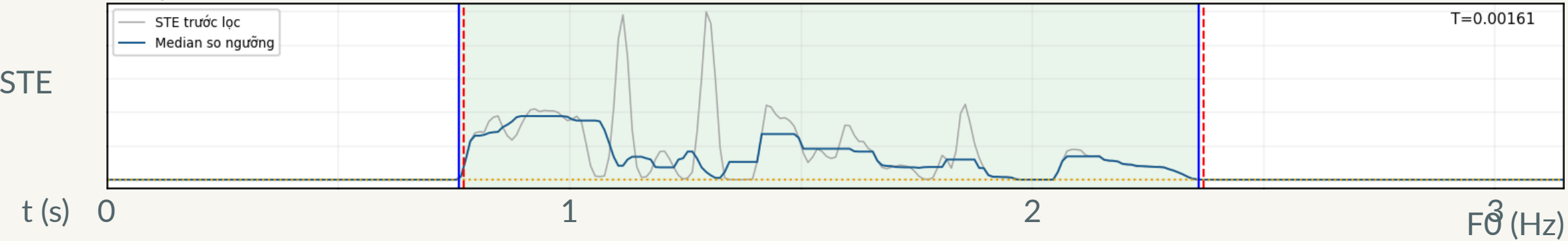
MAE 10 ms

Start 10 ms early. End 10 ms early.



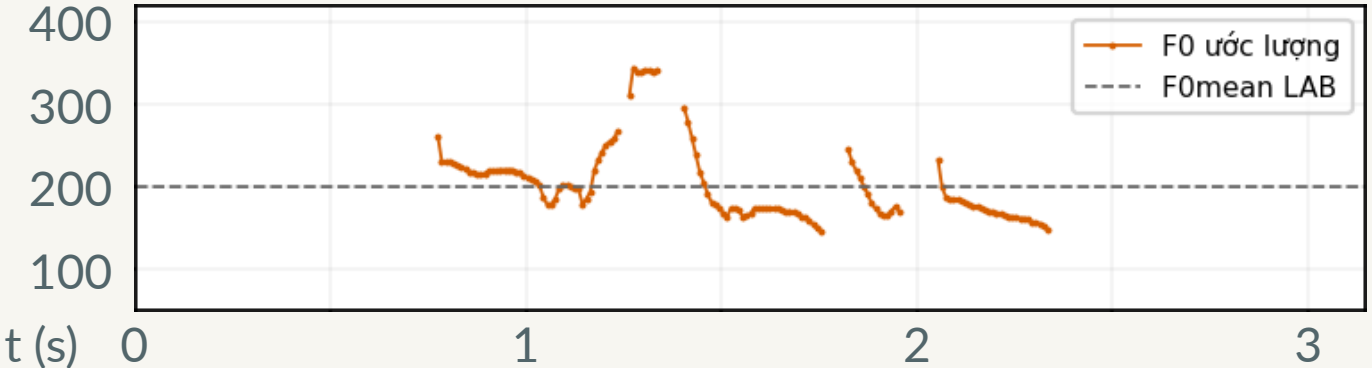
Grey: raw STE. Blue: median STE.

T = 0.0016054242



Energy crosses T before
the LAB transitions

Extra: 0. Missed: 0.



Binary Search: studio_M2

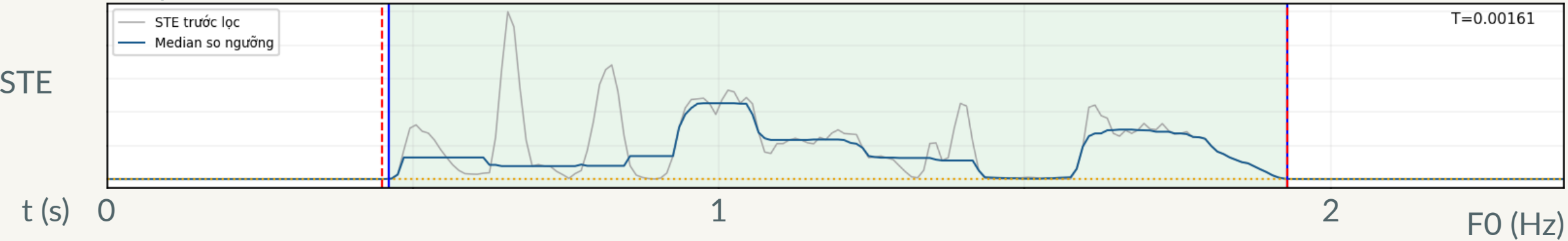
MAE 5 ms

Start 10 ms late. End matches.



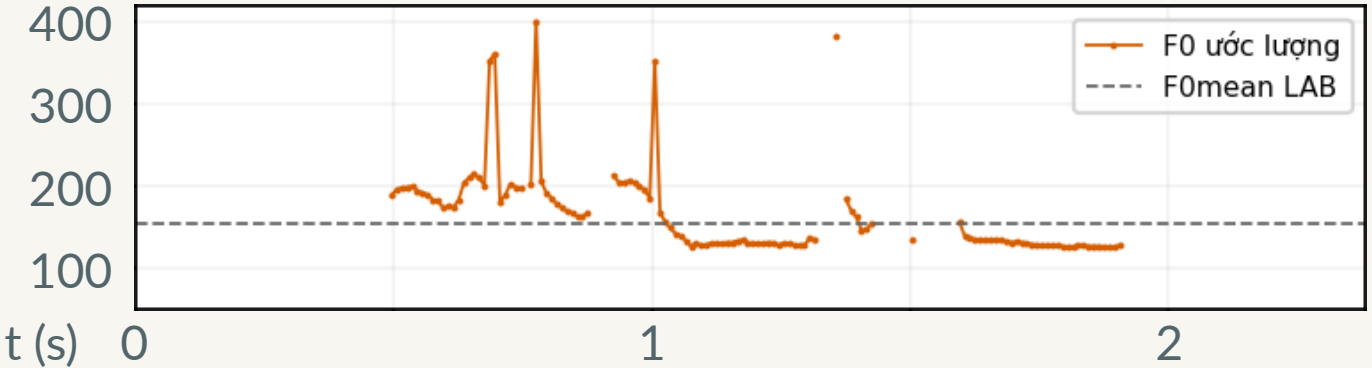
Grey: raw STE. Blue: median STE.

T = 0.0016054242



Possible cause: weak energy near the onset

Extra: 0. Missed: 0.



Histogram

Vương Quốc Trung

TRAIN

TRAIN histograms
128 bins

Try W candidates
Eight values

Score TRAIN boundaries
Errors, then MAE

Selected $W = 30$. Smooth over 3 bins.

TEST

TEST histogram
Normalized STE

First two peaks
Adaptive threshold

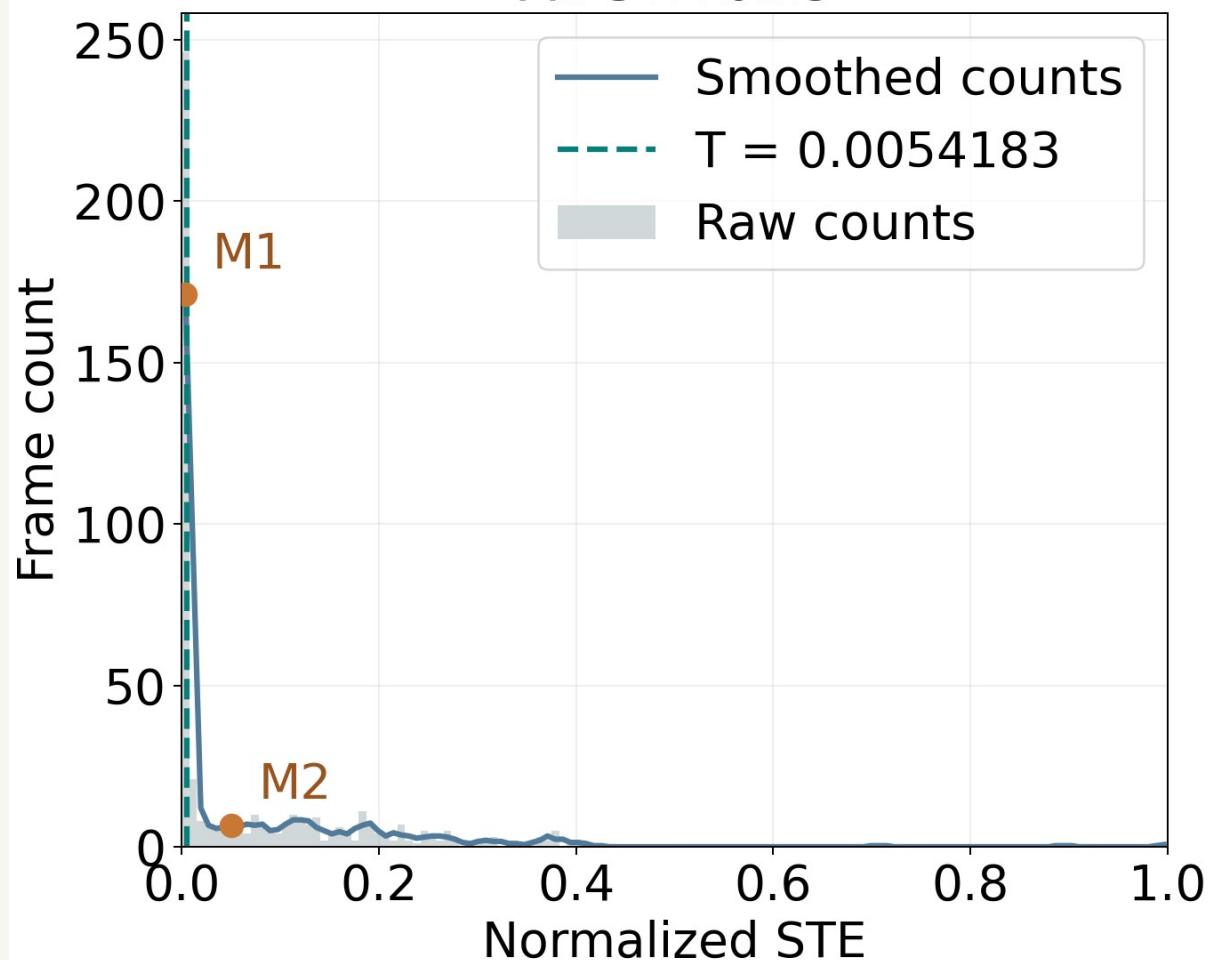
Classify and merge
Silence under 200 ms

Histogram: selected peaks and threshold

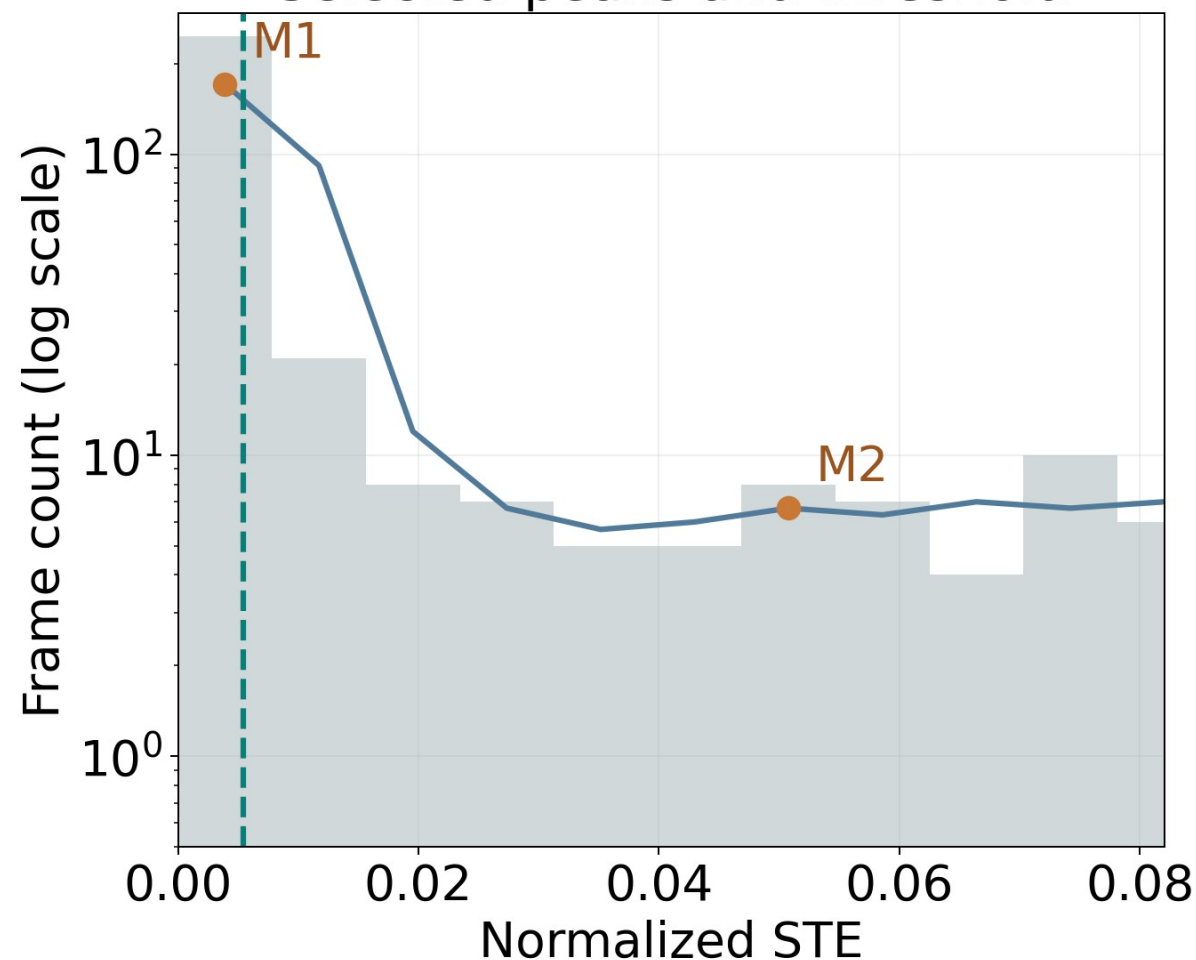
phone_F2. W = 30 from TRAIN.

phone_F2: 128 bins, smoothing 3, W = 30

All STE bins



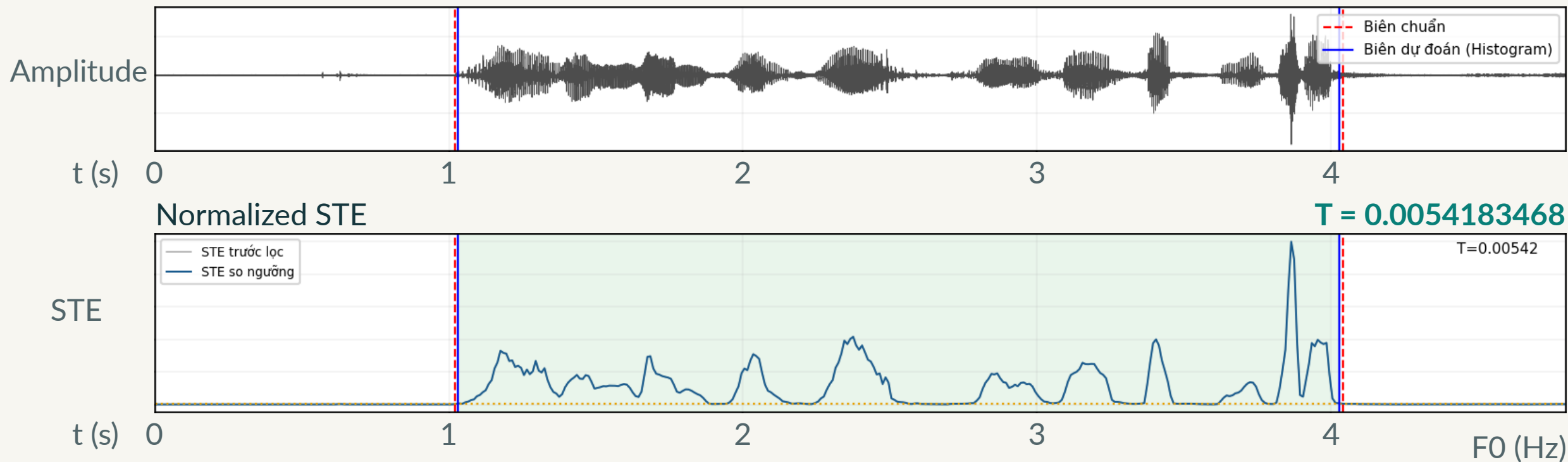
Selected peaks and threshold



Histogram: phone_F2

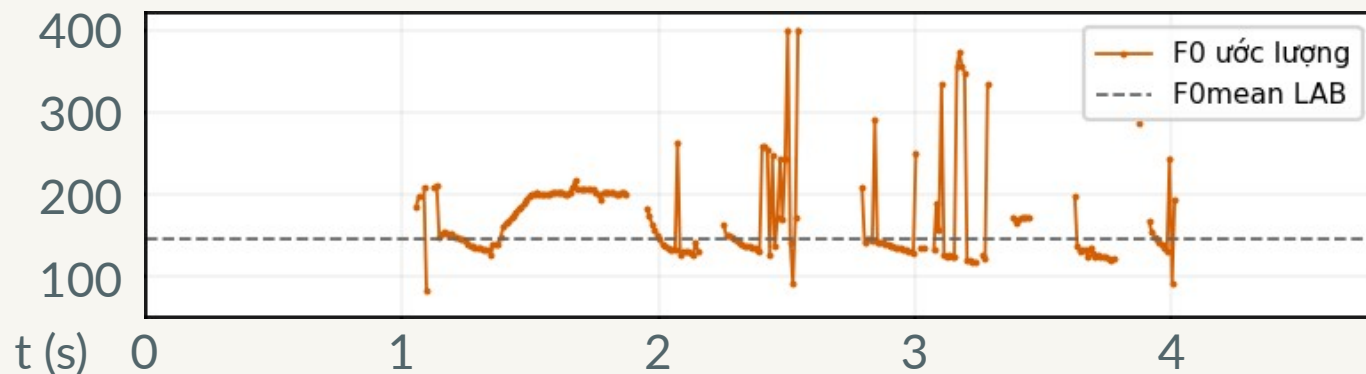
MAE 10 ms

Start 10 ms late. End 10 ms early.



Possible cause: weak energy near the boundaries

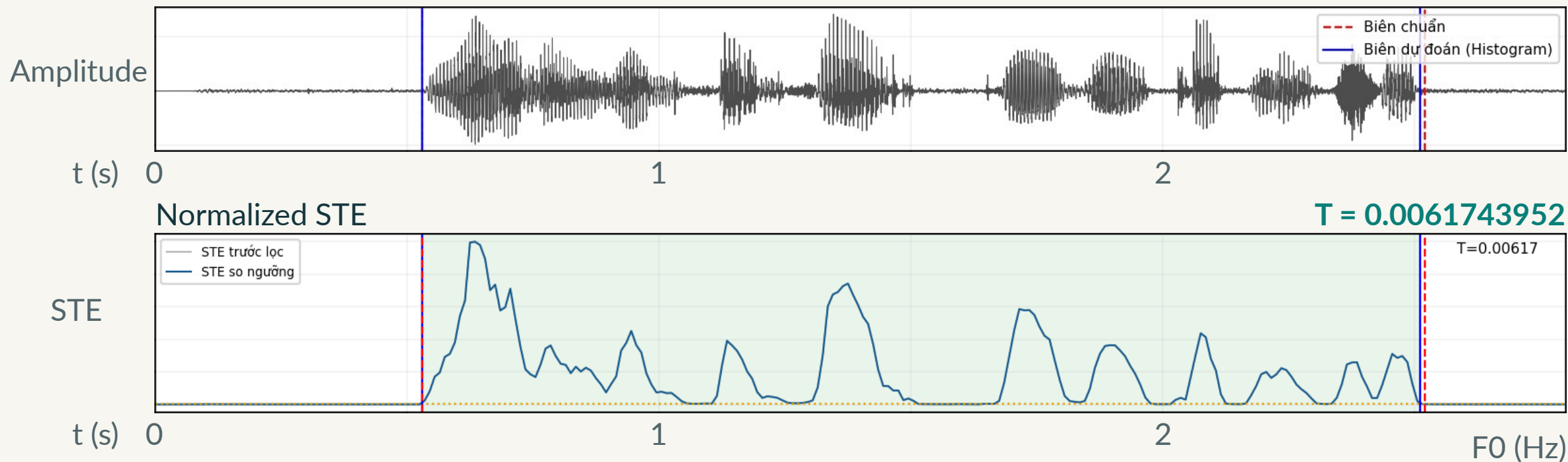
Extra: 0. Missed: 0.



Histogram: phone_M2

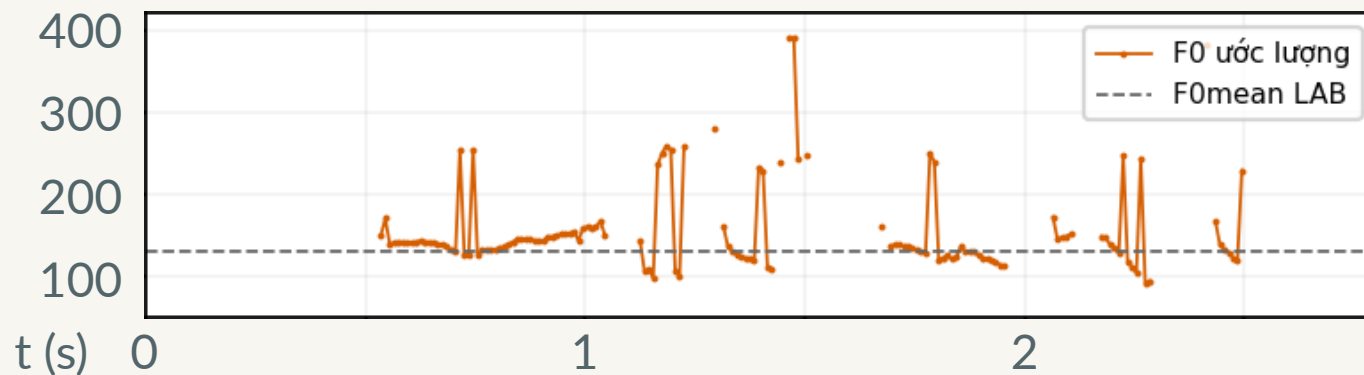
MAE 5 ms

Start matches. End 10 ms early.



Possible cause: weak energy near the boundaries

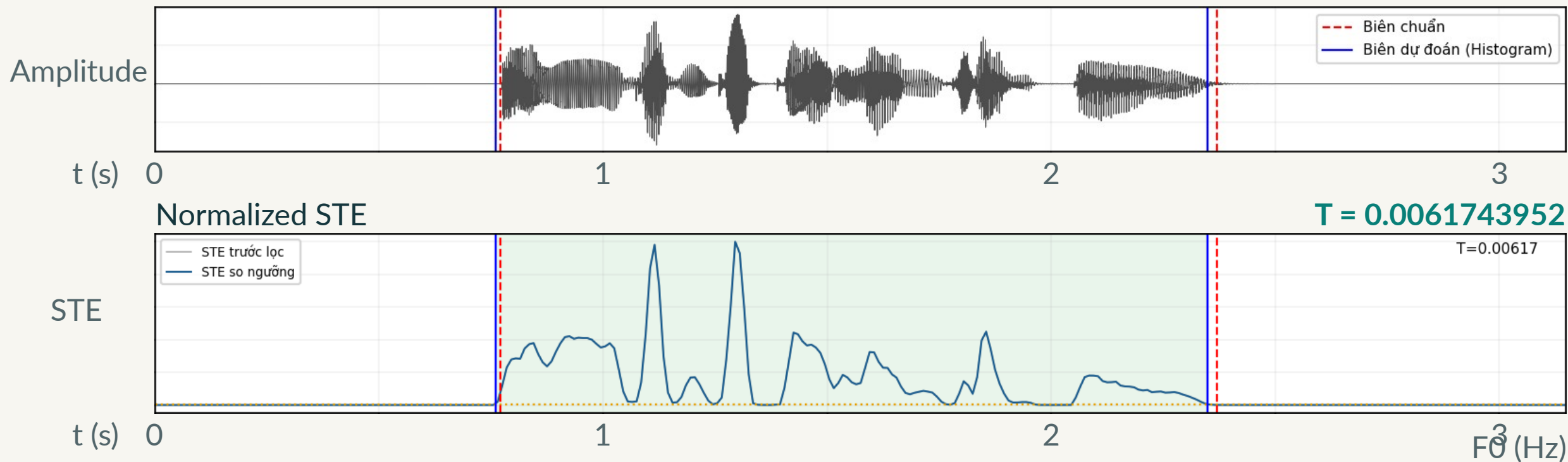
Extra: 0. Missed: 0.



Histogram: studio_F2

MAE 15 ms

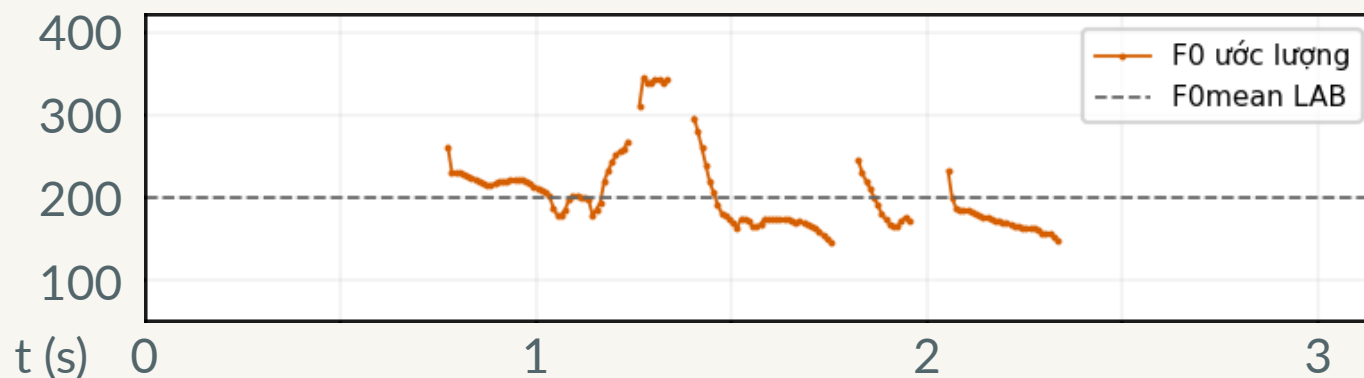
Start 10 ms early. End 20 ms early.



Onset window includes speech.

Tail energy falls below T .

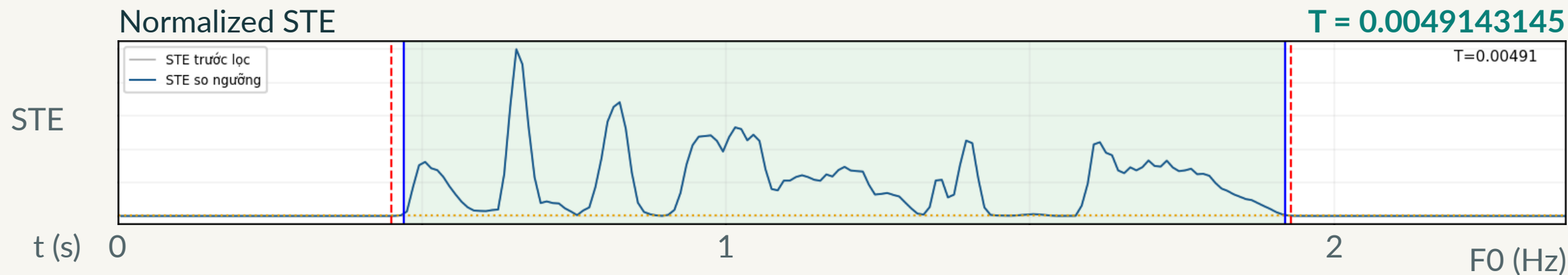
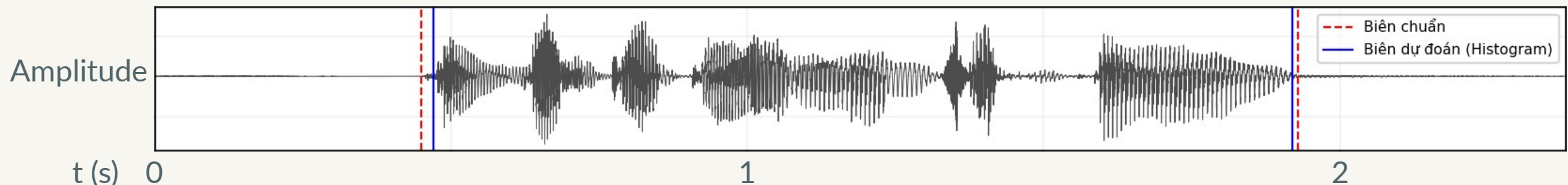
Extra: 0. Missed: 0.



Histogram: studio_M2

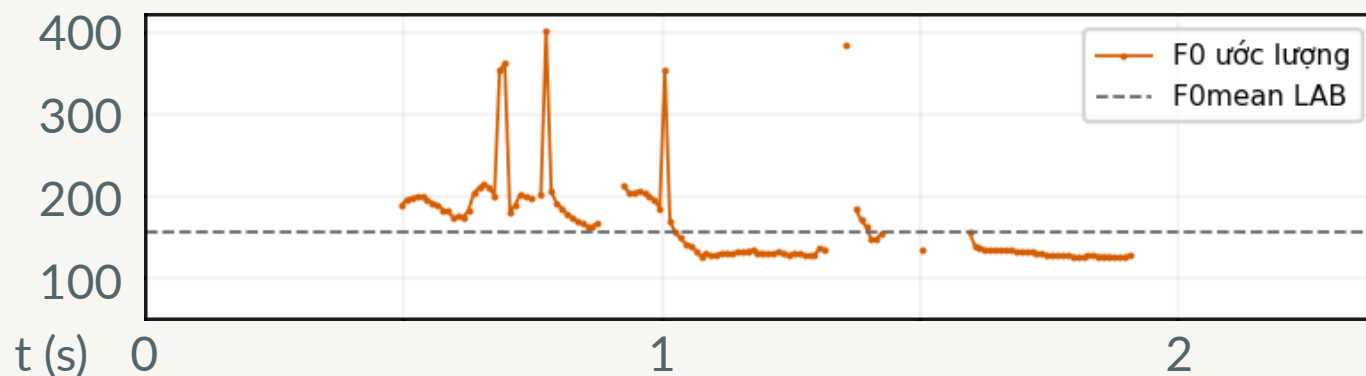
MAE 15 ms

Start 20 ms late. End 10 ms early.



Possible cause: weak
energy near the boundaries

Extra: 0. Missed: 0.



Gaussian Statistics

Đinh Huỳnh Nguyễn Khang

TRAIN



503 silence, 794 speech. $T = 0.0027634166$.

TEST

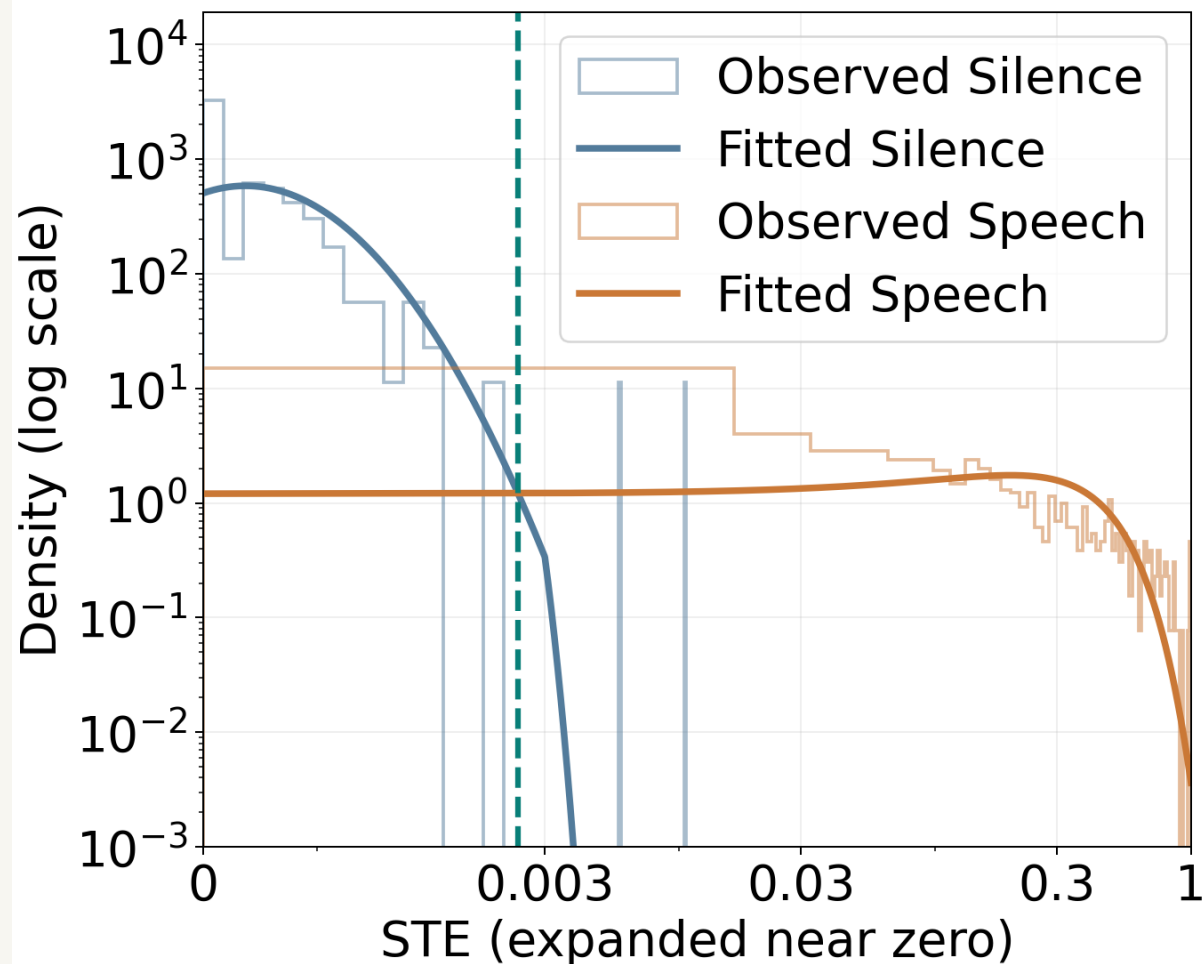


Gaussian: observed data and crossing

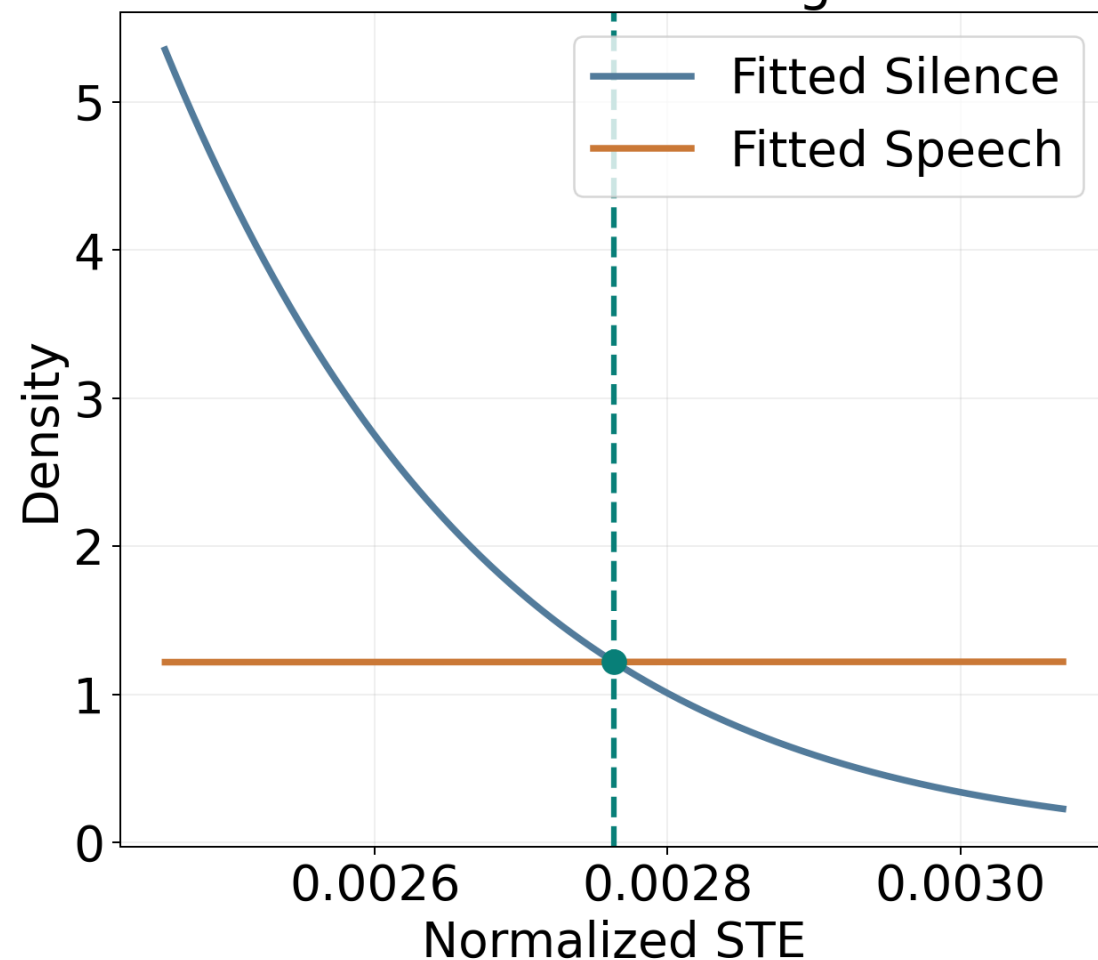
TRAIN estimates. Zoom on the chosen crossing.

Gaussian TRAIN: 503 silence, 794 speech, $T = 0.0027634$

Observed TRAIN and fitted models



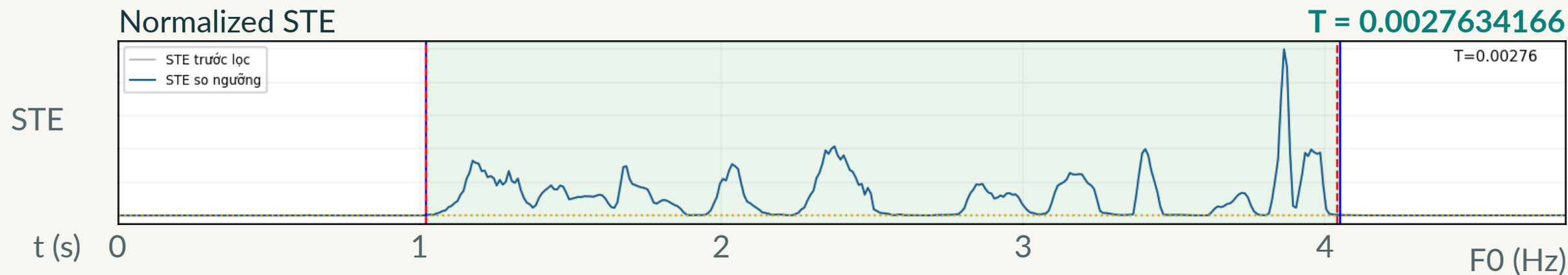
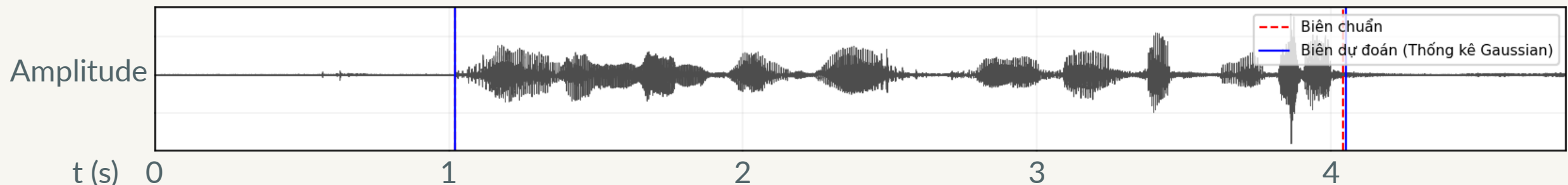
Chosen crossing



Statistics: phone_F2

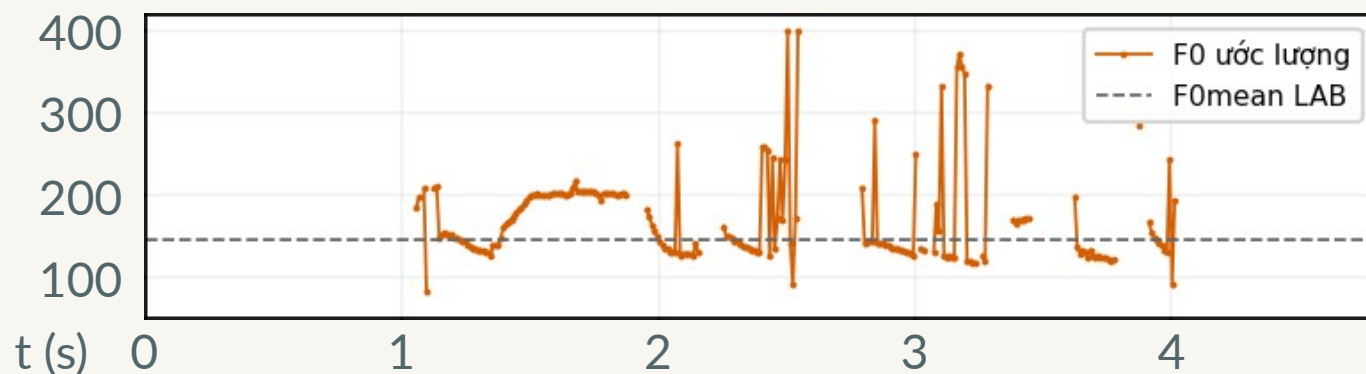
MAE 5 ms

Start matches. End 10 ms late.



Tail STE remains above
the shared threshold

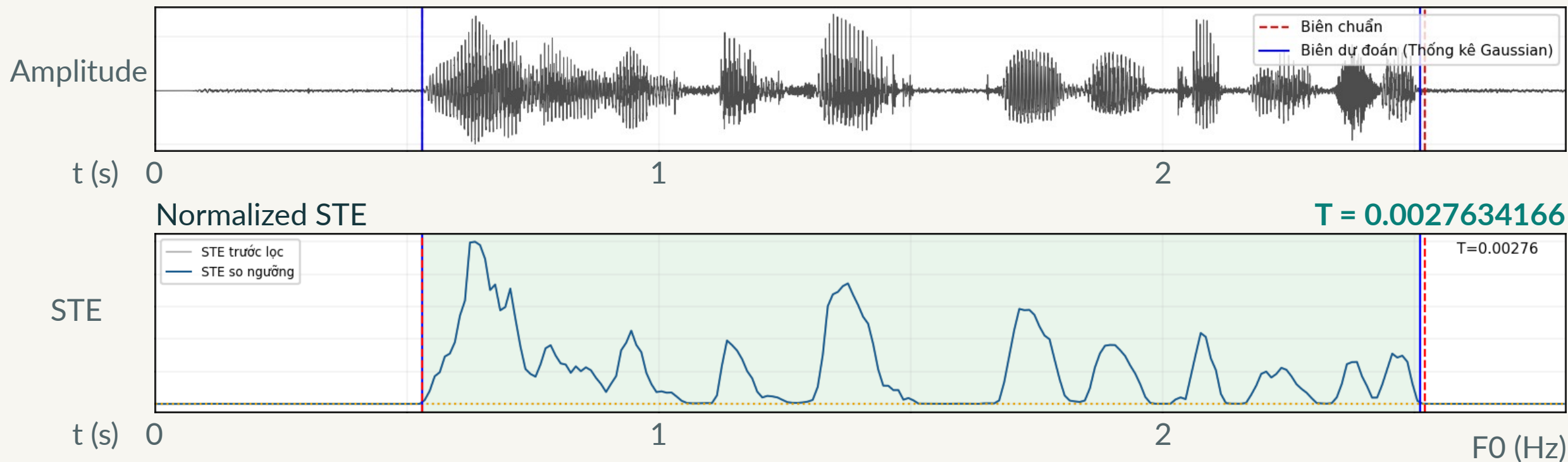
Extra: 0. Missed: 0.



Statistics: phone_M2

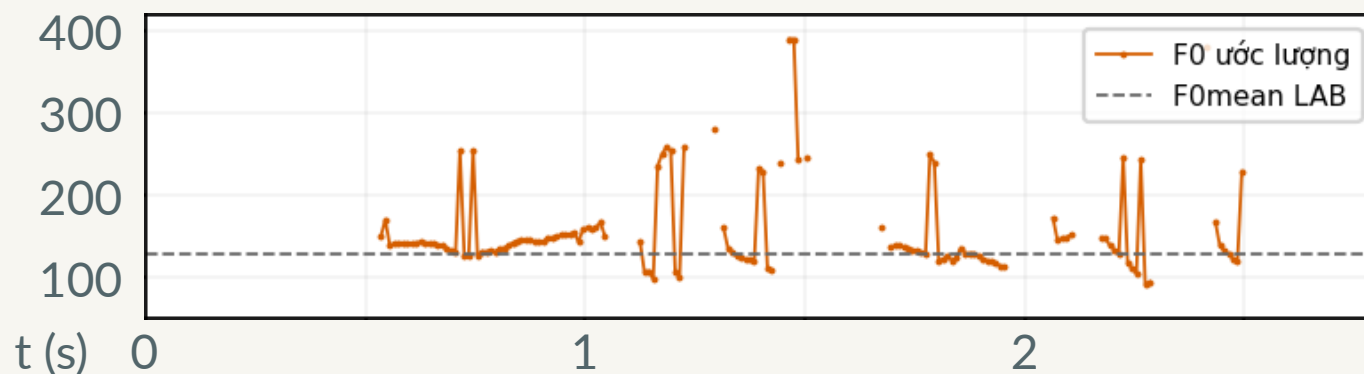
MAE 5 ms

Start matches. End 10 ms early.



Possible cause: weak energy near the boundaries

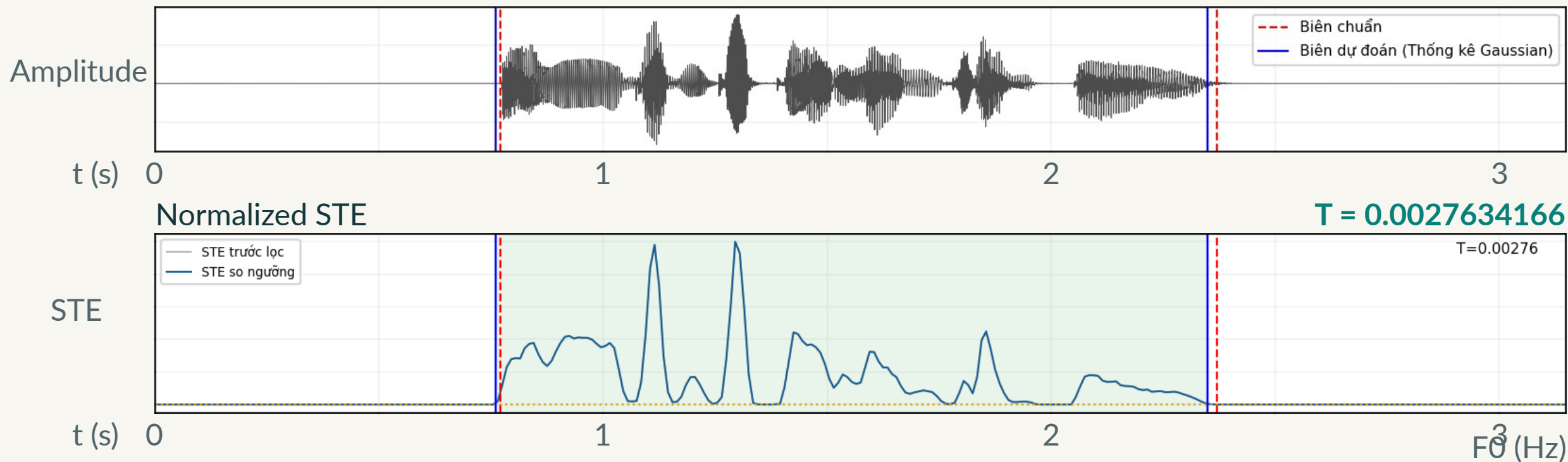
Extra: 0. Missed: 0.



Statistics: studio_F2

MAE 15 ms

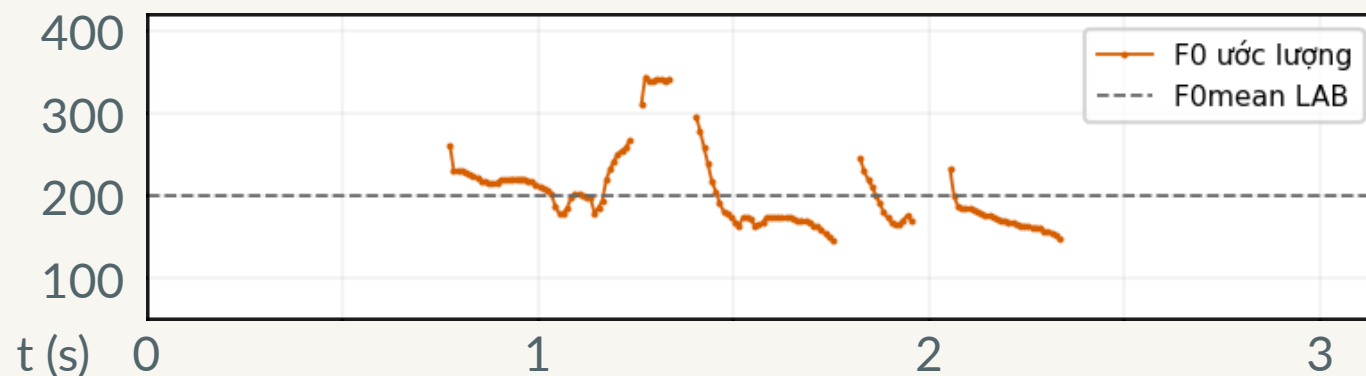
Start 10 ms early. End 20 ms early.



Onset window includes speech.

Tail energy falls below T.

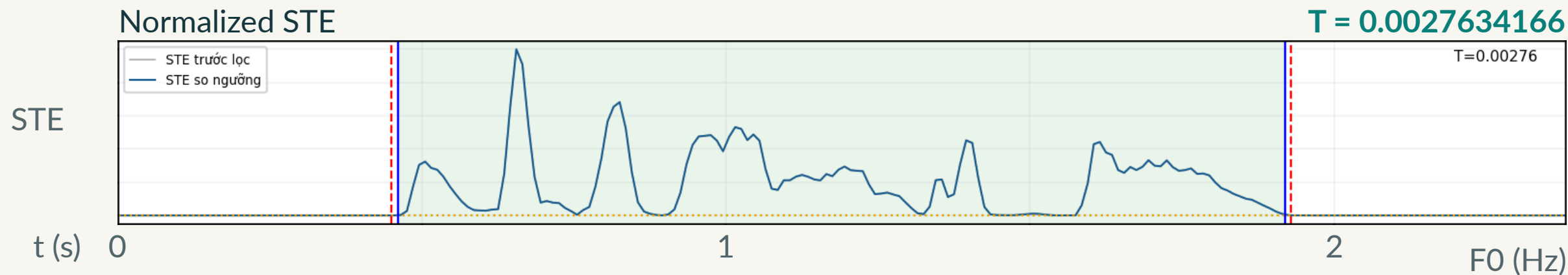
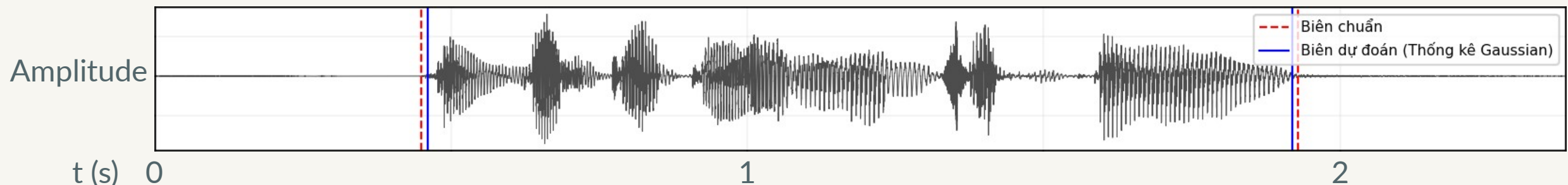
Extra: 0. Missed: 0.



Statistics: studio_M2

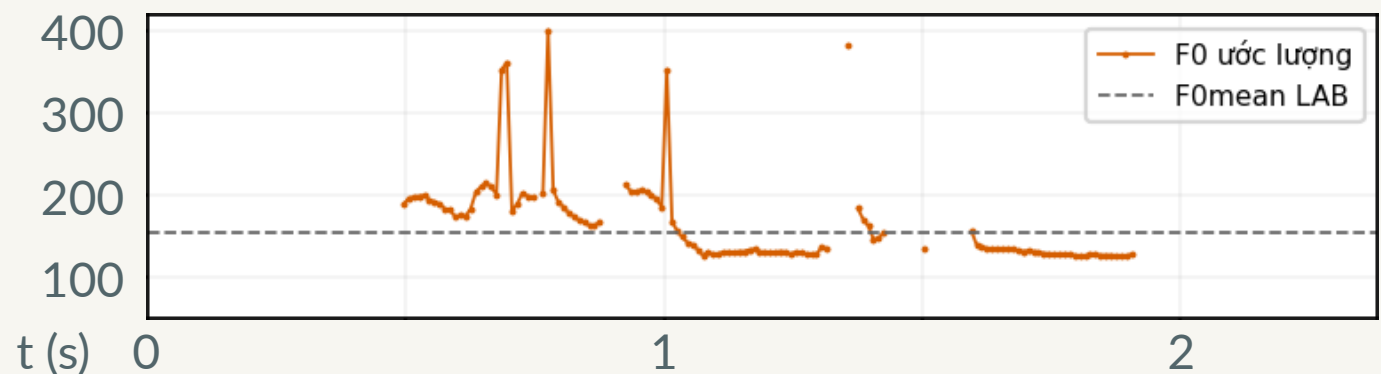
MAE 10 ms

Start 10 ms late. End 10 ms early.



Possible cause: weak
energy near the boundaries

Extra: 0. Missed: 0.



Statistics has the lowest mean TEST MAE (ms)

TEST file	Binary	Histogram	Statistics
phone_F2	30	10	5
phone_M2	0	5	5
studio_F2	10	15	15
studio_M2	5	15	10
Mean	11.25	11.25	8.75

Thank you

Group 03

Võ Thanh Quân

Vương Quốc Trung

Đinh Huỳnh Nguyên Khang